

THE KCP LIMITED

SAFETY THROUGH DESIGN

Introduction

- Over time, the level of safety achieved will relate directly to the caliber of the initial design of facilities, hardware, equipment, tooling operations layout, work environment, and work methods, and their redesign as continuous improvement is sought.
- The goal of continuous redesign and improvement of operations is to reduce the number of errors until operations are as error proof as human effort can make them.

Introduction (Cont.)

- On July 25, 2005, the American National Standards Institute (ANSI) approved a new standard: ANSI/AIHA Z10-2005, *Occupational Health and Safety Management Systems*.
- First time in U.S., a national consensus standard for safety and health management systems applicable to organizations of all sizes and types.
- It is not an OSHA regulation, it is about best practices.

ANSI/AIHA Z-10 Standard Elements

- Safety and health needs are addressed in the design and redesign processes.
- Hazards are identified and analyzed, and risks are assessed and prioritized.
- A prescribed hierarchy of controls is used to reduce risks to an acceptable level.
- A management-of-change procedure is implemented so that hazards and risks are properly considered when changes are made.
- Safety and health specifications are included in purchasing documents and contracts to avoid bringing hazards and risks into the workplace.

NIOSH Workshop

- July 2007 NIOSH Workshop on Prevention through Design (PtD)
- Move design considerations into daily practices
- Practice of safety applying hierarchy of controls in design and redesign aspects
- NIOSH Definition of PtD: *Addressing occupational safety and health needs in the design and redesign processes to prevent or minimize the work-related hazards and risks associated with the construction, manufacture, use, maintenance, and disposal of facilities, materials, and equipment.*

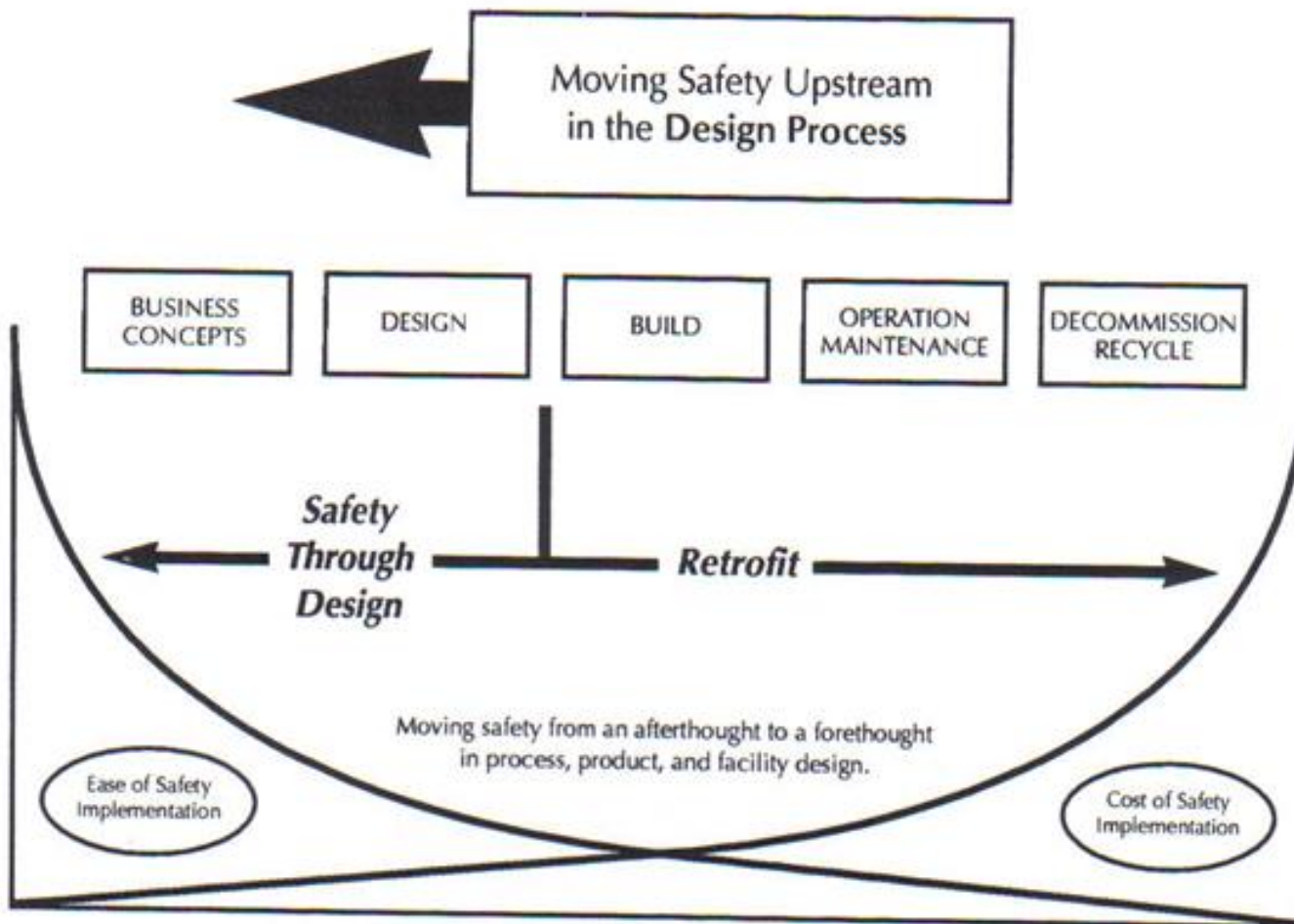
Why Focus on Safety through Design?

- The design stage offers the greatest opportunity and most cost-effective time to anticipate, analyze, eliminate, or control hazards.

What is Safety through Design?

- Safety through design is defined as the integration of hazard analysis and risks assessment methods early in the design and engineering stages and the taking of the actions necessary so that the risks of injury or damage are at an acceptable level.

Design Diagram



Design Diagram

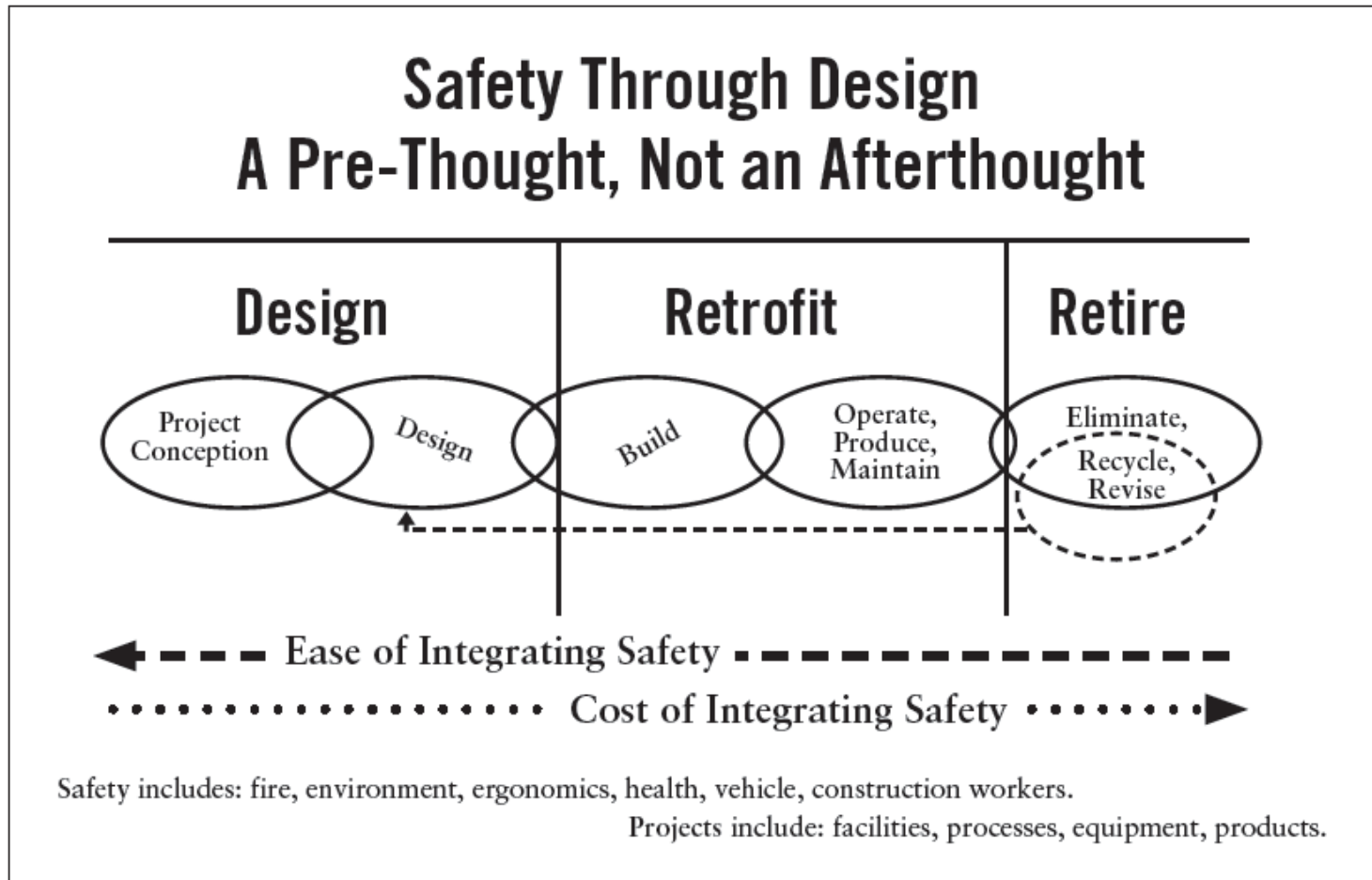


Figure 1-1. The model for safety through design

Benefits of Safety through Design

- Significant reductions in injuries and illnesses, damage to the environment and costs
- Productivity increase
- Reduced operating costs
- Retrofitting costs avoided

Benchmarking

- Benchmarking is the standard of excellence against which other similar things are measured or judged.
- Learn from others experiences.
- Cultural change to accept safety by design rather than the reactive approach.

W. Edwards Deming

- “Improve constantly and forever the system of production and service, to improve quality and productivity, and thus constantly decrease costs.”
- Change quality to safety.

Integrating Safety through Design into the Design Process

- In the design and redesign process, management seeks to avoid, reduce, or eliminate the probability and severity of a hazard potential being realized and causing an incident.

Definitions

- Risk—an estimate of the probability of a hazards-related incident or exposure occurring and the severity of harm or damage that could result.
- Safety—state for which the risks are judged to be acceptable.

Definitions (Cont.)

- Two aspects of risk:
 - Avoiding, eliminating or reducing the *probability* of a hazards-related incident or exposure occurring
 - Minimizing the *severity* of harm or damage, if an incident or exposure occurs

Definitions (Cont.)

- Minimum risk is the goal.
- Minimum risk is achieved when all risks from possible hazards are at a realistic minimum and acceptable.
- Minimum risk does not mean zero risk, which is unattainable.

Definitions (Cont.)

- All risks to which safety through design derives from potential hazards.
- Hazards—the potential for harm: characteristics of things (equipment and dusts) and actions or inactions of individuals

Definitions (Cont.)

- A risk assessment commences with hazard identification and analysis through which the probable severity of harm or damage is established; concludes with an estimate of the probability of the incident or exposure occurring.

Definitions (Cont.)

- ALARP (as low as reasonably possible): the level of risk that can be further lowered only by an increment in resource expenditure that can not be justified by resulting decrement of risk.
- Acceptable risk: the risk for which the probability of a hazard-related incident or exposure occurring and severity of harm or damage resulting are ALARP and tolerable in the setting.

Definitions (Cont.)

- Hierarchy of controls: a systematic process, considering steps in a ranked and sequential order, to choose the most effective means of eliminating or reducing hazards and risks.
- Safety design process:
 - 1) facilities, equipment, operations, energy
 - 2) work methods, procedures, personnel
 - 3) products for human use

Hazard Analysis and Risk Assessment Process

- Establish the analysis parameters
- Identify the hazards
- Consider the failure modes
- Determine the frequency and duration of exposure
- Assess the severity of consequences
- Determine the occurrence probability
- Define the risk
- Rank risks in priority order

Hazard Analysis and Risk Assessment Process (Cont.)

- Develop remediation proposals
- Follow up on actions taken
- Document the results
- Risk matrix provides method to categorize combinations of probability and severity of harm.

Risk Assessment Matrix Example

Table 1–A. Risk Assessment Matrix

<i>Occurrence Probability</i>	<i>Severity of Consequence</i>			
	<i>Catastrophic</i>	<i>Critical</i>	<i>Marginal</i>	<i>Negligible</i>
Frequent	High	High	Serious	Medium
Probable	High	High	Serious	Medium
Occasional	High	Serious	Medium	Low
Remote	Serious	Medium	Medium	Low
Improbable	Medium	Medium	Medium	Low

Integrating Safety through Design

- Companies should apply the following priorities to design and redesign processes:
 - First—Eliminate or reduce risks in design
 - Second—Reduce risks by substituting less hazardous materials
 - Third—Incorporate safety devices
 - Fourth—Provide warning devices
 - Fifth—Administrative controls (methods, training, work schedules)
 - Sixth—Provide and use personal protective equipment

Role of the Safety Practitioner

- The safety professional can influence the design of the workplace and work methods at three critical points:
 - **Preoperational stages**
 - Before building, system, or equipment is operational, a safety professional has the greatest opportunity to identify and analyze hazards.
 - Hazards are avoided, controlled, or eliminated before they result in losses.
 - **Operational stages**
 - After building, equipment, or system is operational, safety measures through continuous improvement process.
 - Anticipate, identify, and evaluate hazards before injuries or deaths occur.
 - **Post-incident stages**
 - After incident has occurred, safety professional must work to improve safety.
 - Investigate, determine casual factors, and implement design changes to correct current hazards.

Proactive versus Reactive Responses

- Research and experience show that a company achieves the greatest effectiveness and economy when dealing with hazards in the preoperational stage of the design processes.
- By taking such a proactive approach, the company obtains designs that can reduce employee risk, improve productivity, and lower unit costs.
 - A proactive company asks safety-related questions while a building, work system, or equipment is being designed.
 - Too often, companies and safety professionals concentrate their safety efforts on worker behavior instead of work methods and procedures.

Alan D. Swain

- “Management should forego the temptation to place the burden of accident prevention on the individual worker.”
- Work situations are created by management.

Alphonse Chapanis

- “The improvement in system performance that can be realized from the design of equipment is usually greater than the gains that can be realized from the selection and training of personnel.”

Dr. William Haddon—Unwanted Energy Transfer

- Quoted “for all injuries and illnesses, an unwanted and harmful transfer of energy or exposure to an injurious environment is a factor.”
- Reduction of damage produced by energy transfer.

Guidelines: Safety through Design

- Companies should consider ergonomics and the strengths and limitations of workers in their design process.

Implementing a Safety through Design Process

- Companies should:
 - Establish clear-cut objectives
 - Assess hazards probability/severity

Hazard Analysis—Risk Assessment

- Establish the analysis parameters—select system, equipment, etc.
- Identify the hazards—cause of incidents
- Consider the failure modes—hazard potential and result in incident
- Describe the exposure—number of people, types, population, and aspects of environment, frequency of incident

Hazard Analysis—Risk Assessment (Cont.)

- Assess the severity of consequences—number fatalities, injuries, illnesses, value property, and environment impact
- Determine the probability of hazard being realized
- Summary conclusion—probability and severity
- Develop proposals to remedy hazards—acceptable risk

Assessing Hazard Probability and Severity

- Probability of a hazard potential being realized and a hazard-related incident occurring is described in probable occurrences per unit of time, events, population, items or activity.
- Quantitative assessment is difficult; more of a qualitative measure.

Implementing a Safety through Design Process (Cont.)

- Companies should:
 - Hazards analysis techniques
 - If hazard analyses and risk assessments are to be effectively made, those assigned the responsibility must be skilled in the use of the special analytical techniques available.
 - Establish design review procedures
 - A few companies have established procedures requiring that hazards be addressed in the design processes for new or altered facilities and equipment.

General Design Safety Process

- Have safety through design checklists
 - An organization's appropriation request for new projects or major alterations may include a proposal review procedure requiring that hazards be properly addressed.
 - If so, having a well-crafted project review checklist makes such a procedure more effective.
 - See General Design Safety Checklist in text.

Management of Change

- The objective is to prevent the introduction of new hazards when changes are made to technology, equipment, facilities, work processes, etc.

Management of Change (Cont.)

- Change analysis ensures that:
 - Hazards resulting from changes have been identified, evaluated, and had corrective measures implemented
 - New hazards are not caused by the change
 - The change does not negatively affect previously resolved hazards
 - The change does not make an existing hazard more severe

Including Safety Specifications in Purchasing Documents

- Prevents introducing hazards and risks into the workplace
 - Have specifications included in purchase orders so suppliers and vendors know your specific requirements
 - Hazards and risks in the workplace are at a practical minimum, significant risk reduction results, and fewer injuries and illnesses will occur